

Algorithms Deep Dive: Recall, Ranking, Typos, Personalization

DFA/Trie Prefix Matching

Purpose: Fast candidate recall based on typed prefixes.

Approach: Store terms in trie/DAWG; each node represents prefix; DFA minimizes states.

Process: Normalize input (case fold, accents removed), walk trie, return candidates.

Complexity: $O(m)$ lookup where m is prefix length.

Strengths: Deterministic, sub-millisecond lookups, compact storage with DAWG.

Use case: High QPS autocomplete queries.

Fuzzy Tolerance

Purpose: Handle typos and misspellings.

Algorithms: Levenshtein automaton (edit distance 1–2), n-gram Jaccard overlap.

Normalization: Lowercasing, accent folding, token cleanup before fuzzy matching.

Constraints: Limit expansion for latency; apply only if prefix recall insufficient.

Strengths: Improves recall in presence of errors, critical for mobile typing.

Semantic Embeddings for Recall

Purpose: Capture meaning-based matches beyond lexical overlap.

Approach: Use embeddings (e.g., word2vec, BERT) to represent entities and queries.

ANN Index: HNSW, IVF, or ScaNN for top-K nearest neighbor search.

Triggers: Long queries, low recall from prefix/fuzzy.

Strengths: Expands results to cover synonyms, concepts, related entities.

Latency: 10–25 ms typical with optimized ANN indexes.

Ranking via Scoring Models

Purpose: Re-rank recalled candidates by relevance.

Signals: Exactness of match, recency of updates, popularity (clicks/selections), personalization score.

Models: Linear combination, gradient boosted trees, or learning-to-rank.

Features: Merge lexical, semantic, and behavioral features.

Outcome: Higher CTR, more relevant top results, explainability through score breakdowns.

Typos Handling via Normalization and Keyboard Distance

Normalization: Lowercasing, accent removal, Unicode canonicalization.

Keyboard Distance: Model typos based on proximity of keys (e.g., 's' near 'a').

Algorithms: Damerau-Levenshtein distance accounts for transpositions.

Pipeline: Normalize → calculate keyboard-aware edit distance → filter candidates.

Outcome: Improves matching quality for fast/mobile typing errors.

Personalization with Strict Opt-In and Residency Rules

Purpose: Improve ranking with user-specific context while preserving privacy.

Signals: Recent selections, affinity scores per entity type, session history.

Opt-In: Users must explicitly consent; anonymized identifiers used.

Residency: User data stays in its originating region for compliance (GDPR/CCPA).

TTL: User features expire after fixed windows (e.g., 30 days).

Outcome: Higher relevance for repeat users, no violation of privacy guarantees.